

SCOTT POWERS

saberpowers.github.io

ACADEMIC EXPERIENCE

Rice University	2023–
Assistant Professor, Sport Analytics, (by courtesy) Statistics	2023–
Director, Hutchinson Leadership Initiative in Sport Analytics	2026–

SPORTS EXPERIENCE

FULL-TIME

Houston Astros	2022
Assistant General Manager	

Los Angeles Dodgers	2017–2021
Director, Quantitative Analysis	2019–2021
Senior Analyst, Research & Development	2017–2018

PART-TIME/CONSULTING

Teamworks (formerly Zelus Analytics)	2020, 2023–2024
Staff Data Scientist	2023–2024
Senior Data Scientist	2020

AZ Alkmaar	2015–2020
Data Analyst	

Oakland Athletics	2015–2016
Analytics Consultant	

EDUCATION

Stanford University	2017
Ph.D. Statistics	
Advisor: Rob Tibshirani	

University of Chicago	2013
M.S. Statistics	

University of North Carolina at Chapel Hill	2011
B.S. Mathematical Decision Sciences and Mathematics	
<i>Highest Honors from the Department and Highest Distinction from the University</i>	

RESEARCH

WORKING PAPERS

Powers, Ramani S, Hahn J, Schaefer AJ
“Lead distance under a pickoff limit in Major League Baseball: A sequential game model”

arXiv:2601.15608

Zhu R, Love D, **Powers**

“Ball path curvature and in-game free throw shooting proficiency in the National Basketball Association”

arXiv:2507.01238

Millington E, **Powers**

“A comps-based approach for interpreting tree-based predictions with an application to the NFL draft”

Powers, Iglesias V

“Pitch trajectory density estimation for predicting future outcomes”

JOURNAL ARTICLES

Powers, Yurko R (2026)

“Swinging, Fast and Slow: Interpreting variation in baseball swing tracking metrics”
to appear in *The American Statistician*

Powers, Stancil L, Consiglio N (2025)

“Estimating individual contributions to team success in women’s college volleyball”
Journal of Quantitative Analysis in Sports **21**(2) 117–135

Burton T, **Powers**, Burns C, Conway G, Leach F, Senecal K (2023)

“A data-driven greenhouse gas emission rate analysis for vehicle comparisons”
SAE International Journal of Electrified Vehicles **12**(1)

Powers, McGuire V, Bernstein L, Canchola AJ, Whittemore AS (2019) “Evaluating disease prediction models using a cohort whose covariate distribution differs from that of the target population”

Statistical Methods in Medical Research **28**(1) 309–320

Powers, Hastie T, Tibshirani R (2018)

“Nuclear penalized multinomial regression with an application to predicting at bat outcomes in baseball”
Statistical Modelling **18**(5–6) 388–410

Powers, Qian J, Jung K, Schuler A, Shah NH, Hastie T, Tibshirani R (2018)

“Some methods for heterogeneous treatment effect estimation in high dimensions”
Statistics in Medicine **37**(11) 1767–1787

McGinnis L, **Powers**, Bangs D, Cherry A, Tibshirani R, Natkunam Y (2016)

“Double-hit diffuse large B-cell lymphomas with MYC gene rearrangements more commonly involve BCL2 than BCL6 gene rearrangements as the second hit: A large scale single institution study”
Laboratory Investigation **96** 362A

Burton T, **Powers** (2015)

“A linear model for estimating optimal service fraction in volleyball”
Journal of Quantitative Analysis in Sports **11** 117–129

Powers, Hastie T, Tibshirani R (2015)

“Customized training with an application to mass spectrometric imaging of cancer tissue”
The Annals of Applied Statistics **9** 1709–1725

Powers, DeJongh M, Best AA, Tintle NL (2015)

“Cautions about the reliability of pairwise gene correlations based on expression data”
Frontiers in Microbiology **6** 650

Powers, Gopalakrishnan S, Tintle NL (2011)

“Assessing the impact of non-differential genotyping errors on rare variant tests of association”
Human Heredity **72** 153–160

Saavedra S, **Powers**, McCotter T, Porter MA, Mucha PJ (2010)

“Mutually-antagonistic interactions in baseball networks”
Physica A: Statistical Mechanics and its Applications **389** 1131–1141

INVITED CONFERENCE TALKS

“Winning baseball games by solving statistics puzzles”

International Conference on Statistics and Data Science

Vancouver 2025

“Estimating individual contributions to team success in women’s college volleyball”

MIT Sloan Sports Analytics Conference

Boston 2024

New England Symposium on Statistics in Sports

Boston 2023

INVITED DEPARTMENT SEMINARS

“Winning baseball games by solving statistics puzzles”

Northeastern University, Network Science Institute

Mar 2026

Texas A&M University, Institute of Data Science

Feb 2026

Southern Methodist University, Department of Statistics and Data Science

Sep 2025

Johns Hopkins University, Department of Electrical and Computer Engineering

Sep 2025

“Player evaluation based on game performance”

New York Knicks (National Basketball Association)

Nov 2024

TSG 1899 Hoffenheim (Bundesliga)

Apr 2024

“Nuclear penalized multinomial regression”

Rice University, Department of Statistics

Apr 2023

INVITED CONFERENCE PANELS

“Building the Academic Playbook for Sports Analytics Education”

MIT Sloan Sports Analytics Conference

Boston 2026

“The Past, Present, and Future of Sports Analytics in Industry and the Academy”

Joint Statistical Meetings

Nashville 2025

“Data Science and Analytics in Sports Performance and Management”

Columbia Symposium on AI and Sports

New York 2024

CONTRIBUTED CONFERENCE TALKS

“Swinging, Fast and Slow: Untangling intention and timing error from bat speed and swing length”

Cascadia Symposium on Statistics in Sports

Vancouver 2024

Saberseminar

Chicago 2024

“Pitch trajectory density estimation for predicting future outcomes”

Conference of Texas Statisticians

Houston 2024

Saberseminar

Chicago 2023

“Jointly predicting exit velocity and launch angle for batter-pitcher matchups” Saberseminar	Boston 2016
“Nuclear penalized multinomial regression for predicting at bat outcomes in baseball” Joint Statistical Meetings	Chicago 2016
“True wOBA: Estimation of true talent level for batters” SABR Analytics Conference	Phoenix 2016
“Rewarding batters for baserunner advancement: A ridge-regressed Rasch model” Saberseminar	Boston 2015

STUDENT CONFERENCE TALKS

“A comps-based approach for interpreting tree-based predictions with an application to the NFL draft” Elisabeth Millington @ Carnegie Mellon Sports Analytics Conference	Pittsburgh 2025
“Ball path curvature and in-game free throw shooting proficiency in the National Basketball Association” Ruoqian (Judy) Zhu @ New England Symposium on Statistics in Sports	Boston 2025
“The Two-Foot Rule: A game theoretic analysis of the pickoff limit in Major League Baseball” Jacob Hahn @ Cascadia Symposium on Statistics in Sports Jacob Hahn @ Saberseminar	Vancouver 2024 Chicago 2024
“Modifying k -means clustering to optimize positioning in volleyball” Naomi Consiglio @ Women in Sports Data Symposium	Philadelphia 2024
“Do we learn more about AAA batters when they face better pitches?” Jeff Brover @ Saberseminar	Chicago 2024
“Time warping for clustering pitcher deliveries” Drew Haugen @ Saberseminar	Chicago 2024

STUDENT CONFERENCE POSTERS

“A comps-based approach for interpreting tree-based predictions with an application to the NFL draft” Elisabeth Millington @ New England Symposium on Statistics in Sports	Boston 2025
“A statistical approach to sport climbing difficulty and progression” Elizabeth Sepúlveda @ Cascadia Symposium on Statistics in Sports	Vancouver 2024
“Not All Features Are Created Equal: Player clustering and evaluation” Andrew Kang @ Opta Forum	London 2024

FUNDING

EXTERNAL

Major League Baseball Research Gift

“Identifying biomechanical risk factors for pitching injuries” PI: S Powers, \$100,000	2024–2026
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INTERNAL

Houston Methodist-Rice University Center for Human Performance Seed Grant Program

“Utilizing Ultrasound-Based Shear Wave Elastography to Assess UCL Changes in Collegiate Pitchers”

PIs: S Powers, R Jack, \$46,483

2026

Rice University Creative Ventures Conference and Workshop Development

“Rice Soccer Analytics Conference” PI: S Powers, \$10,000

2024–2025

TEACHING

* indicates a course I designed

Rice University

Instructor, *SMGT 430: Introduction to Sport Analytics

Spring 2024, 2026; Fall 2024, 2025

Instructor, *SMGT 435: Baseball Analytics

Spring 2024; Fall 2024, 2025

Instructor, STAT 699: Topics in Statistical Sciences

Spring 2026; Fall 2024, 2025

Instructor, *SMGT 432: Soccer Analytics

Fall 2023

Instructor, *SMGT 434: Football Analytics

Fall 2025

Instructor, SMGT 373: Sport Analytics Internship

Spring 2026

Stanford University

Instructor, *STATS 50: Mathematics of Sports

Spring 2016

TA, STATS 305A: Applied Statistics I

Fall 2016

TA, STATS 216: Introduction to Statistical Learning

Winter 2014, Summer 2015, Winter 2017

TA, STATS 202: Data Mining and Analysis

Fall 2013, Summer 2014

TA, STATS 50: Mathematics of Sports

Fall 2014

University of Chicago

Instructor, STAT 23400: Statistical Models and Methods

Spring 2013

TA, STAT 22000: Statistical Methods and Applications

Fall 2011, Spring 2012, Fall 2012

MENTORSHIP

Rice University

Principal Investigator, Sport Analytics Undergraduate Research Lab

2024–

Students/Alumni: Jeff Brover, Naomi Consiglio, Lucca Ferraz, Jacob Hahn, Rahul Herrero, Elisabeth Millington, Zach Pool, Elizabeth Sepúlveda, Luke Stancil, Lou Zhou, Ruoqian (Judy) Zhu

Coordinator, Sport Analytics Graduate Reading Group

2023–2024

Coordinator, Big Data Bowl Support Group

2023

SERVICE

To the Profession

Lead Organizer, American Soccer Insights Summit

2024–

Sports Column Co-Editor, *CHANCE*

2026–

Associate Editor, SCORE Network

2024–

Organizer, ASA Virtual Sports Analytics Conference

2026

Rice University

Faculty Associate, Wiess College

2024–

Instructor, STaRT@Rice (School of Social Sciences)

2024, 2025

Instructor, Owl Days Classroom Sampler (Office of Admission)

2024

Speaker, Special Occasion Dinner (Friends of Fondren Library)	2024
Speaker, Brains in a Bar (Association of Rice Alumni)	2024
Member, University Committee for Faculty and Staff Benefits	2024

Peer Review

Journal of Quantitative Analysis in Sports (8), *Journal of Sports Analytics* (2), *Journal of Sport Management*, *Orthopaedic Journal of Sports Medicine*, *International Journal of Sports Science & Coaching*, *Journal of Sports Sciences*, *Experimental Physiology*, *Computer Methods in Biomechanics and Biomedical Engineering*, *Proceedings of the National Academy of Sciences*

PROFESSIONAL DEVELOPMENT

Rice Center for Teaching Excellence

INSTILL Mini-Grant Program	2023–2025
Reading Group	2023–2025
Symposium on Teaching and Learning	2024
Workshop on Inclusive Teaching Strategies	2023

Conferences Attended

Women in Sports Data Symposium	Philadelphia 2023, 2024
Minority Trailblazers in Sports Conference	Houston 2023

AWARDS

Graduate Research Fellowship, National Science Foundation	2011
McCormick Fellowship, University of Chicago	2011
Mathematical Decision Sciences Award, UNC Department of Statistics & Operations Research	2011
Student Speaker Award, Pi Mu Epsilon National Meetings	2010
National Merit Scholarship, National Merit Scholarship Corporation	2007
Jack Kavanagh Memorial Youth Baseball Research Award, SABR	2005, 2006, 2007

ACTIVITIES

Sports Analytics Club, Stanford University	2014–2017
Co-President	2015–2016
Intercollegiate Club Baseball, Stanford University	2013–2015
Intercollegiate Club Volleyball, Stanford University	2013–2016
Intercollegiate Club Volleyball, University of Chicago	2011–2013
Intercollegiate Club Volleyball, University of North Carolina at Chapel Hill	2007–2011
President	2008–2010